Shifting the Evidentiary Burden: A Roadmap for Verifiable Exclusion from AI Training Datasets

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When a company claims that an individual's data was not used to train a model, or that the data has been removed, nobody outside the company can verify the claim. The person who objects (an artist, an author, a researcher, or a data subject) must show that their data is inside a training set they cannot see, while the company, the only one able to inspect that set, need only assert that it is not. Courts and regulators facing such disputes have three inadequate options: accept the company's word, compel disclosure of proprietary and often personal training data, or order destruction of data and models, whose execution again rests on the company's word. We propose a fourth: a verifiable refusal scheme, in which a model owner proves, rather than asserts, that refused data was not used to train its model, without disclosing its training data. We provide the guarantees such a scheme must provide and give a construction, FENCE (Framework for Exclusion via Non-membership Cryptographic Evidence), which is built from any universal cryptographic accumulator with three standard properties. The model owner publishes a short commitment to its training data and, together with an attestation that the model was trained on the committed data, can prove to any refuser that every item on their refusal list is absent from the model, while revealing nothing else. FENCE supports both proactive refusal, before a model is trained, and retroactive removal, after which the model was retrained. The main cost of our construction falls on the model owner, while verifying proofs is cheap.

On the legal side, we map how data exclusion and removal are negotiated across different regimes, noting how the mere existence of an exclusion proof may change legal arguments able to be made. We study two hypothetical FENCE deployments across different jurisdictions, concluding with recommendations on its legal operation—including a proposed rebuttable presumption rule that motivates AI providers to comply with a verifiable exclusion scheme by shifting evidentiary burden if they don't.

1 Introduction

Amazon told parents they could delete their children's Alexa voice recordings. According to the US Federal Trade Commission (FTC), it kept them anyway, and nobody outside the company could tell. In May 2023, the American Federal Trade Commission ordered Amazon to pay 25 million in damages for unlawful data retention and appropriation. The FTC claimed that Amazon had violated the Children's Online Privacy Protection Act (COPPA) and its own public-facing assurances by retaining child voice recordings indefinitely–and in some cases, ignoring parents' deletion requests. In a court order, Amazon was ordered to delete all violating data, and prohibited from using data associated with a deletion request to train its speech algorithms [44].

An order itself, though, does not independently verify deletion. In Amazon's case, no audit of deletion was demanded. Instead, for the next ten years, an vice president overseeing Alexa was required to produce annual certification that she was personally unaware of noncompliance, with her attestation supported by bookkeeping and document production upon request. Ultimately, compliance with data deletion still rested on Amazon's word—the same word it had gone against when it knowingly retained the voice records.

Among legal instruments requiring AI companies to delete data, evidentiary gaps persist when an assertion is made about *data provenance* or *remedial data exclusion*. Lacking a cheap, private approach to checking for noncompliance, ex-post revelation of noncompliance has mostly happened by chance. Illustrative cases include *Reddit v. Anthropic*, where Anthropic claimed it had ceased to scrape Reddit's forums for model training, but Reddit later pointed to server logs recording 100,000 accesses[40]; Everalbum's usage of consumer photos to train face-recognition models after

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promising deletion, where later compliance with a model-deletion order was confirmed only by the company's own report [25]; and the case of Cambridge Analytica—preceding the commercial boom of neural networks but nevertheless setting precedent—where the consulting firm continued using non-consensually harvested Facebook data to improve user-targeted political advertisement algorithms, even after it confirmed the data was destroyed [24]. In an EU case, the French data protection authority ordered Clearview AI to delete data on individuals in France, and justify compliance within two months. The company submitted no proof, and was fined €5.2 million [29].

These disputes differ in law, in parties, and in outcome, but they share a structure. When a user objects to the use of its data for training, the company can answer that the data was not used, that its use has ceased, or that it has been removed in the form of a certification, an affidavit, or compliance report. However, this leaves the regulators, courts, and user themselves with no way to verify the company's claim. The second option, which is an extreme case, is when regulators order a destruction of the data and of the models the owner trained on, as the Federal Trade Commission (FTC) did in the case with Everalbum [25]. This option still relies on the company's claim with no evidence–how do we truly know they deleted the correct model and training data?

The third option is for a court or regulator to compel disclosure of the training data, through discovery, an audit, or a transparency mandate. This can verify the model owner's word, but only by giving an auditor full access to the training data and to how the model was trained, which exposes trade secrets and, in many cases, other people's personal data. Refusers, moreover, must still trust the auditor and receive nothing they can verify themselves. This option is an instance of what Bamberger et al. call a *verification dilemma*: a fact needs to be verified, and the obvious way to verify it discloses sensitive information beyond the fact itself [4]. The dilemma is not tied to a specific regime or jurisdiction, and in disputes over an AI model's training set it has left a consistent gap in auditing and enforcement, whether exclusion is claimed by the model owner or ordered by a judge. A verifiable exclusion scheme, which we introduce below, avoids this dilemma: it lets the model owner prove exclusion without disclosing its training data, and it places the burden of proof on the model owner.

The lack of adequate options for resolving these disputes motivates our contribution: a fourth option for courts, regulators, and contractual counterparties, made possible by cryptography. We call it a verifiable exclusion scheme. In such a scheme, the model owner must actively prove, rather than merely assert, that a user's data was not used to train its model, and must do so without disclosing its training data. We then give a concrete construction of such a scheme, which we call FENCE. We consider two cases in which a user may refuse the use of their data. In the proactive case, the user's request is made before the model is trained, so that once the model is trained, the model owner can prove that the user's data was not included. In the retroactive case, removal is requested after the model has been trained, so the model must be retrained without the data, and the model owner must then prove that the new model was trained without it. Once such a scheme exists and can be required by policymakers, a company can no longer answer a request to demonstrate exclusion by saying that no demonstration is possible. Alongside the scheme and the FENCE construction, we analyze its place in the law and how it would fit into existing legal frameworks, evaluating its deployment in two legal landscapes: the EU's opt-out regime for AI training and FTC deletion orders in the US (Section 5).

1.1 Contributions

Our contributions intertwine cryptography and law. Our legal contributions are motivated by specific properties of our cryptographic scheme, and each cryptographic design choice answers a question that the law leaves open.

- **A verifiable exclusion scheme.** In Section 3, we define what it means for a model owner to prove that data items were excluded from its training data, and state the guarantees that such a scheme must provide to be in line with our legal analysis.
- **A map of how verifiable refusal enters data law.** Across copyright law, privacy law, and remedial data deletion regimes, we provide a cross-jurisdiction view of legal instruments where *proactive* and *retroactive* data exclusion are negotiated. We outline how verifiable refusal closes an evidentiary gap, and argue that the mere possibility of cryptographic verifiability is enough to shift the regulatory landscape (Section 2.2).
- **A cryptographic construction of a verifiable exclusion scheme.** We propose FENCE (a Framework for Exclusion via Non-membership Cryptographic Evidence), which uses cryptographic building blocks to construct a verifiable exclusion scheme (Section 4). We explain why FENCE provides each of the scheme's guarantees, reducing each one to a standard property of its building blocks.
- **Two deployment case studies with different legal levers.** In the EU, we illustrate how FENCE's *proactive refusal* capability (Section 4.6) complements the AI Act's Article 53(1)(c) obligations, meets the needs of sectoral stakeholders, and extends the capabilities of a distributed opt-out registry that the Commission is considering implementing. In the US, where a comprehensive privacy law does not exist, we show how an FTC order (*FTC v. Amazon*) may periodically leverage FENCE's *retroactive refusal* capability to cheaply and verifiably audit deletion compliance (Section 5).
- **A rule shifting evidentiary burden.** Within the EU opt-out regime, we propose a paired *rebuttable presumption of use* and *rebuttable presumption of compliance* that incentivize model providers to proactively comply with a verifiable exclusion scheme. We outline how this redistributes the burden of proof so it does not lie unfairly with the harmed party, and argue that applied cryptography makes these paired rules especially viable (Section 5).

Why both disciplines are needed. A purely technical proposal would go unused as a company that can already answer objections with its word has no reason to adopt a protocol that comes with cost and time to verify its word. A purely legal proposal is also limited: a rule that demands a proof of exclusion either puts privacy at risk for everyone in the training set, collides with trade secrecy, or only mandates the company's certification without verifying the claim. For this reason we propose FENCE in a paper that designs the mechanisms and rules together.

To our knowledge, this is the first work to define verifiable exclusion from AI training data as a problem in its own right, stating what any scheme must guarantee independently of how it is built, and to pair such a scheme with legal mechanisms that put it to use. Our scheme gives evidence to anyone who refuses their data, including people who never supplied data to the model owner, collectives such as artists' guilds, collecting societies, or the users of a platform, and whole sources such as websites or online accounts. On the legal side, we start from the imbalance of power between model owners and refusers, and argue that once such proofs are feasible, the burden of production can fairly be moved to the model owner, the only party able to carry it.

1.2 Paper Outline

Section 2 establishes the legal problem and reviews related work, including machine unlearning. Section 3 defines a verifiable exclusion scheme, including the setting, the parties involved, and the guarantees it must provide. Section 4 presents our construction, FENCE, and explains why it provides each guarantee, reducing each one to a standard property of its building blocks. Section 5 offers cross-regime perspectives on how FENCE may be integrated into law. The appendices give the

formal syntax and definitions of a cryptographic accumulator (Appendix A), the formal definitions for our verifiable exclusion scheme (Appendix B), and the asymptotic costs of FENCE (Appendix C).

2 Background

2.1 Information asymmetry and burden of production

When a party accuses an AI company of using their data to train a model, the *burden of proof* refers to the standard they must meet to have the fact legally established in court. In most cases, the *burden of proof* is construed of two parts: the *burden of production* and the *burden of persuasion.* By default, both fall on the party making the claim. *Burden of production* decides whose responsibility it is to produce enough initial evidence that the fact is taken seriously in court rather than preemptively dismissed. *Burden of persuasion* decides which party is responsible for proving the claim.

In the accusation described above, the claimant, bearing burden of production by default, must do the difficult job of demonstrating their data is inside a training set they cannot see, while the model owner, withholding data for privacy or proprietary reasons, need only assert that it is not. Output regurgitation—which is rare and inconsistent—is among few types of confirmable evidence.

Some regimes, in recognizing the power asymmetry tied up in evidentiary duty, have enacted interventions. Generally, burden of production can be reversed by statutes. In the US courts, the civil discovery period also allows parties to request evidence from each other. In the EU, the GDPR accountability principles stated in Article 5(2) mandate that data controllers must not only comply with data protection principles but also proactively demonstrate that compliance. The EU Product Liability Directive (2024/2853) recognizes the information asymmetry between customers harmed by defective products and companies manufacturing them, and makes noncompliance with mandatory safety regulations a legal trigger for a rebuttable presumption of defect. This means the company, not the plaintiff, must provide evidence of its products' non-defect.

However, even if model providers were charged with the duty of demonstrating compliance, the charge is meaningless if there is no means of demonstration, and as we have seen with current disputes, for trained models there is none in use.

2.2 Regimes of data exclusion

To better illustrate how our verifiable exclusion scheme—and our construction FENCE—fits into law, we provide a brief, cross-jurisdiction view of two regimes that audit or compel data exclusion from a model's training set: *copyright law* and *privacy law.* In each process, we outline the theories of harm and rectifying actions that have tended to emerge when data misappropriation is contended. We then briefly argue how the existence of a FENCE-like system—able to verify *ex-ante* claims that infringing data has not been used, as well as *ex-post* claims that infringing data has been destroyed—closes an auditing gap, and in some contexts, shifts the legal landscape by giving the mere act of proof evasion new evidentiary meaning.

Our examples bias towards the US and the EU not because we believe their legal landscapes are most deserving of study, but because the paper's later case studies—arising from and limited by our own expertise—are drawn from these landscapes.

2.2.1 AI models and copyright: verifying non-use claims. The World Intellectual Property Organization defines copyright as "the rights that creators have over their literary and artistic works" [45]. These generally apply to expression and not underlying idea, and may be split into *economic rights* (rightholders have certain exclusive rights, such as over reproduction and distribution) and *moral rights* (non-economic rights such as that of recognized authorship).

In legal disputes concerning generative AI's consumption of intellectual property, theories of infringement have been argued at three points along the training process.

- **The training set.** In both the US and EU, companies possessing training sets containing reproductions of unlawfully accessed copyrighted works have been found liable. This was the case in *Bartz v. Anthropic* (Northern District of California) [42] and *GEMA v. Suno* (Munich) [36], where copyrighted works were pirated. In *Kneschke v LAION* [34], the Hamburg District Court did *not* rule the non-profit providing the LAION training set liable, as works were accessed lawfully, and for research purposes—a use-case protected by a EU copyright reform.
- **Memorization in model weights.** In *GEMA v. OpenAI* [35], a Munich court found that memorization of artwork in a model's weights, inferred by identical regurgitation in output, constitutes an infringing copy. In US courts, this is currently unsettled.
- **Infringing outputs.** In the case of non-verbatim outputs, theories of infringement have argued that outputs which reproduce protectable expression (characters, plot, setting, literary voice), even non-verbatim, can be found liable. In a consolidated lawsuit against OpenAI, authors argued infringing expression, and the claim survived dismissal in a U.S. federal court [43]. In the Chinese *Ultraman* case [28], the Guangzhou Internet Court found an AI platform liable for producing images substantially similar to the copyrighted Ultraman character after Ultraman-related prompts.

Within copyright, FENCE's proof of training set exclusion comes in most directly as a means for companies to demonstrate, without revealing training data, that they did not infringe at the dataset creation step. If the source data has not even been ingested, then memorization in weights is impossible, and infringing outputs are highly improbable, meaning a FENCE-style attestation is a stronger guarantee than needed to defend against the two subsequent theories, but it nevertheless functions as one. Cryptography allows ex-ante non-use claims to be *verified*.

It does the same for ex-ante claims, offering public proof that a model owner has complied with a data deletion request, and future model releases no longer contain infringing data.

2.2.2 AI models and privacy: verifying lawful acquisition and retention practices . Privacy constitutes a more expansive, regime-diverse umbrella than copyright. While in copyright law, a harmed party is construed as an individual whose authored work has been misappropriated, *privacy law* construes the harmed party as an individual whose *personal* data has been misappropriated. We narrow our focus to discrete and individually attributable records—such as voice recordings, face images, biometric templates, and medical records—since their presence in a training set can be tested record-by-record.

Generally, these records are protected in the EU via the GDPR. In particular, Article (6) mandates that any processing of personal data—extending to collection—must have lawful basis, and the regulation also subjects biometric data to special regulations. The US lacks a federal lawful-basis requirement, and liability is instead found through state and sectoral statutes requiring consent for specific data types.

We identify the following theories of harm that rest on training set claims:

- **Unlawful acquisition.** Data about an individual is gathered non-consensually, through misleadingly acquired consent, or otherwise unlawfully. Prior disputes involving biometric data include lawsuits from multiple EU states and the Illinois ACLU against Clearview AI, which created a training set of face prints derived from social media images. In the Illinois lawsuit, plaintiffs invoked Illinois' Biometric Information Privacy Act (BIPA). [2] In 2021, the FTC settled with Everalbum over allegations that it intentionally used deceptive collection practices to amass training data. [25]
- **Unlawful re-purposing.** Data obtained lawfully, such as for hospital or research use, is appropriated beyond original privacy and consent terms for AI training. The EU (and UK) regulates this through the purpose limitation principle [21]; in 2015, a London hospital trust

transferred 1.6 million patients' records to Google DeepMind without patients' knowledge or consent, and in 2017, the ICO concluded the re-purposing unlawful for reasons of disclosure and consent. [30]

- **Unlawful retention.** Data is retained beyond a lawful duration. This concerns the GDPR's storage limitation principle, as well as the right to erasure and objection. In the US, retention is governed by different regimes according to location and sector. Beyond the BIPA, the federal Children's Online Privacy Protection Act (COPPA) bans retention of children's personal data—including biometric—beyond the stated need, demanding that companies produce a retention policy with a deletion timeline. *FTC v. Amazon* is an example of a violation that resulted in deletion orders. [44]

Within this landscape, claims of ex-ante data exclusion support companies' arguments of non-liability, while evidence of ex-post data deletion is important in verifying that companies have engaged in rectifying behavior after liability is found. In the past, these claims have been substantiated by written testimony and company-provided evidence or situationally negotiated third-party audit schemes that are limited and publicly opaque due to the sensitivity of training data. Cryptography has the potential of making ex-ante and ex-post claims publicly verifiable while still preserving privacy.

2.3 Related work

Verifiable unlearning and proofs of training. The work closest to ours is verifiable unlearning [18], which defines what it means for a server to prove that it has deleted a user's data and retrained, and instantiates the definition with succinct proofs of the training computation for small models. More generally, proofs of training let a model owner show that a model results from training on committed data, through recorded checkpoints [31], zero-knowledge proofs [1, 27], or protocols for verifying training data [11]. Our scheme differs in what it proves and for whom: verifiable unlearning concerns data that was in the training set and is later removed, whereas our scheme also covers proactive refusal, where data never enters the training set. It also separates two layers: the proof of training is produced once per model, by any of the routes above, while each refuser receives a short Proof of Exclusion that it can check cheaply and that reveals nothing else about the training set. Finally, these works focus on the technical guarantee and do not examine how such proofs would operate in law. We design our scheme together with the legal mechanisms that would give it force: its guarantees are chosen to match what a court or regulator needs, and we show how they support rules such as a rebuttable presumption of use (Section 5).

Machine unlearning and disgorgement. Machine unlearning aims to remove the influence of specific data from a trained model without retraining from scratch [9, 32]. Exact methods yield the same model as retraining, while approximate methods only make the model behave similarly, and recent work argues that unlearning cannot deliver what policymakers often expect of it [12]. The corresponding legal remedy is algorithmic disgorgement, the destruction of models trained on unlawfully obtained data. Neither gives an outside party evidence it can check; our scheme does, and lets refusal take effect before training, so that nothing has to be unlearned.

Inference from the model. A refuser could instead try to infer from the model whether their data was used, through membership inference [46, 49] or by extracting memorized training data. Such evidence is probabilistic and often inconclusive: membership inference cannot prove that a model was trained on a given item [48]. Moreover, legal obligations typically concern whether data was used in training, a fact about the training process rather than about the weights. This is the verification dilemma of Bamberger et al. [4] in another form: whether data was used cannot be established from the outside, and establishing it from the inside requires disclosure.

Cryptography in CS&Law. FENCE belongs to a line of work that uses cryptography to reconcile secrecy with oversight. Zero-knowledge proofs can let a defendant test hidden investigative software [6], cryptographic commitments can make algorithms and secret processes accountable [26, 33], and secure multiparty computation can let parties report confidential misconduct settlements, as in Murmurs [10]. Like Murmurs, FENCE pairs a cryptographic protocol with the legal provisions that would put it to use.

3 A Verifiable Exclusion Scheme for AI Training Data

This section introduces our problem setting, the parties involved, and the guarantees that a verifiable exclusion scheme for AI training data must provide.

The setting. We first state what it means for a model to have been trained without a person's data. A model owner has access to many data items that it may use for training its model, which we refer to as the training set. Some of the items in this training set may be items that the model owner may not be permitted to use. The model owner trains a model on a subset of the items found in the training set by using a training algorithm. We also treat training as deterministic once its randomness is fixed, so that the training set, initial weights, and randomness together determine the model. Additionally, the randomness must not be chosen by the model owner, and instead by some public source that the owner cannot influence or read before it has committed its training data. A refuser is a person or organization that holds some data items and wants to know whether the model was trained without them. To decide whether a training item counts as a use of a refused item, we compare their features: publicly agreed properties of an item, such as a passage of its text, what an image looks like, or the website or account it came from. Which features are compared is fixed publicly in advance, before any model is trained, so that neither the model owner nor a refuser can later change what counts as using an item. The training set uses a refused item if some item in it has a feature in common with the refused item, for example if it contains the same passage, an image that looks the same, or content from the same website. We say that a model was trained without the refused items if it was produced by the training algorithm on a training set that does not use any of them.

3.1 A verifiable exclusion scheme

The parties. Besides model owners and refusers, a verifiable exclusion scheme involves three other roles. A *registry* collects refusals on behalf of refusers and keeps the resulting refusal list; it could be, for example, a data-protection authority, a collecting society, or a platform acting for its users, and a refuser may also act on their own without one. A *setup authority*, such as a standards body or a regulator, fixes the public rules of the scheme once, in advance: which features are compared, how they are represented, and the public parameters of the cryptography. It must not keep any secret that would let a model owner cheat, and it plays no role after setup. Finally, a *verifier* is anyone who checks a proof: the refuser, a registry, a regulator, or a court.

A scheme for a verifiable exclusion proceeds in a fixed order, and goes as follows. First, the setup authority fixes the public rules and parameters of the scheme, and then the model owner publishes a description of its model: a short commitment to the agreed features of its training data, a commitment to its training algorithm and to any model it starts from, a fingerprint (hash) of its weights, and a proof that this description is correct. Here a commitment is a basic cryptographic tool that works like a sealed envelope: once published, it fixes what is inside, so the model owner cannot later change it, but it reveals nothing about the contents. The training algorithm and initial weights need not be public, but they must be committed to in this way, so that the owner cannot later claim to have used a different way of training the data. This proof shows that the model was

really trained on the committed data with the committed algorithm and public randomness. A refuser may register a refusal at any time, before or after training, but a proof is always requested for a model whose description has already been published. If the refuser registers the refusal before training, this represents our proactive case, where the refuser asks that their data not be used in the first place. The alternative is the retroactive case, where the refuser registers the refusal after a model has already been trained on their data and asks that it be removed; the proof then concerns a new model trained without those items. In both cases, the proof that the model owner returns after a refuser asks about their items is what we call a Proof of Exclusion.

The refuser computes the representation of the refused items themselves based on the publicly agreed features, so the model owner can never decide how a refused item is represented. The refuser, or anyone else who obtains the proof, then verifies it against the published model description. This description is fixed before any refuser makes a request, so the model owner cannot answer different refusers on the basis of different training sets. Additionally, to protect the model owner's training set and the privacy of other users' data in this set, the Proof of Exclusion must not tell a refuser anything else about the training set beyond whether its own refused items were used.

Requirements. A verifiable exclusion scheme must provide three guarantees: completeness, soundness, and training-set privacy. We describe each one informally below and include more formal definitions in Appendix B.

(1) The first requirement, *completeness*, ensures that a model owner whose training set does not use a refuser's items can always produce a proof that is accepted.
(2) The second requirement, *soundness*, says that if a proof is accepted, then the published model was trained without the refused items. One cannot tell from a model's weights, or from its description, what it was trained on, so this guarantee cannot be stated by looking at the model alone. Instead, we require that whenever a model owner publishes a valid description, it must know a training set that explains the model: training on this set with the committed algorithm, the committed initial weights, and the public randomness reproduces the published model exactly. (In cryptographic terms, this training set can be *extracted* from the model owner.) Soundness then requires that every proof the owner later has accepted is consistent with this one training set, meaning that the training set does not use any of the refused items.
(3) Lastly, *training-set privacy* protects the privacy of the model owner's training set, and ensures a Proof of Exclusion reveals nothing about the training set beyond whether it uses the refused items or not.

Requirements and the law. These requirements are in line with the legal analysis of Section 5. Completeness means that an owner that respects refusals can always demonstrate it, which is what makes a presumption of compliance fair. Soundness means that an accepted proof is evidence that refused data was not used to train the published model. Training-set privacy means that demonstrating compliance does not require disclosing trade secrets or other people's personal data.

4 FENCE System

This section introduces our protocol, FENCE, that satisfies our verifiable exclusion scheme for AI training data. First, we introduce the three building blocks of FENCE, which are refusal tokens (Section 4.1), cryptographic accumulators (Section 4.2), and a way for the model owner to prove their model description is correct (Section 4.3). Using these building blocks, we describe the FENCE protocol (Section 4.4), and state what it guarantees. We then describe how FENCE can be used before and after training for our two separate refusal scenarios (retroactive and proactive), and briefly discuss its asymptotic costs (Section 4.7). The name of our protocol, FENCE, is analogous

to the problem that it solves: just like a fence, a published commitment marks a boundary that everyone can see and anyone can check is intact, without seeing what lies on the other side.

4.1 Refusal Tokens

A refusal (or refusal list) has to take a form that both the refuser and the model owner compute identically, and the refuser must never need to see the training set. FENCE accomplishes this with tokens (Definition 4.1). A token is a representation of one feature of a data item, and two items share a token exactly when they share that feature. Since models are trained on various types of data, which feature is used varies with the deployed use case. To compute tokens, FENCE uses a cryptographic hash function, which turns data of any size into a short fingerprint of fixed length. The same input always yields the same fingerprint, so the refuser and the model owner compute matching tokens without exchanging any data, and different inputs practically never yield the same fingerprint. This implies that matching tokens mean the same feature, and a fingerprint does not reveal the data it came from (except for someone that already holds the item, since they can check whether its token is on a list). Because every token is equally short, refusal lists stay small however large the refused items are.

Definition 4.1 (Refusal token). Let H be a hash function into the accumulator's domain $\mathcal{D}_{\mathsf{pp}}$, and let $\mathsf{Feat}(d)$ denote the set of features of a data item d. The tokens of d are $\mathsf{Tok}(d) = \{H(f) : f \in \mathsf{Feat}(d)\}$. A refusal token is a token $r \in \mathcal{D}_{\mathsf{pp}}$ that a refuser registers on a refusal list; it refuses every item d with $r \in \mathsf{Tok}(d)$.

For visual work, say, a refuser does not want a model owner to use their images. In the simplest case the feature is the image itself, and the tokens are hashes of the images. However, in this case, a model owner could then alter a single pixel, or make some other change small enough to not affect the output of the hash function, and obtain a different token. To make this hard, the feature can instead be a perceptual hash of the image, which is designed so that images that look alike receive the same value and thus the same token [39, 47]. Another type of refusal concerns text or code, where the refuser does not want the model to use their written work. Here the tokens can be computed from passages of a certain length, so that each token represents a chunk of the whole work. As in the image example, the feature can be a locality-sensitive hash, under which two passages that differ in only a few words receive the same value with high probability [15, 16]. Yet another type is a source token, which records where the model owner obtained an item. This can be an identifier such as a domain name or an account on a named service. A source token essentially says "anything from here". This may be one of the most common forms of refusal in practice: a site reserves its content, or a user of a service objects to the use of their account's data.

4.2 Cryptographic Accumulators

FENCE is built from a cryptographic accumulator which is a way to commit to a set of information and later prove that given elements are either in it or not (we defer the formal definition to Appendix A). An accumulator compresses a set of any size into a single short value, which works like a sealed envelope: it is published, it fixes the set (or contents), and it reveals nothing about what the set (or inside) contains. The party that computed it keeps auxilary information that lets it later produce two kinds of short proofs against the published value. This includes a membership proof that shows whether an element is in the set, and a non-membership proof, which shows that every element of a given list is absent from it. Anyone that holds the published value can check either kind of proof, without learning anything about the set itself. An accumulator that supports both kinds of proofs is called universal, and one that is computed once from a fixed set is called

static. For FENCE, a static accumulator suffices, because a model owner commits to a new training set by honoring refusals for each training run.

FENCE relies on three properties of the accumulator. *Correctness* says that the scheme works for honest parties: if the accumulator was computed honestly from a set of data, then every element of the set has a membership proof that verifies, and every list of elements outside the set has a non-membership proof that verifies. In FENCE, correctness is what lets a model owner whose training set does not overlap with the refusal list always produce an accepted proof. *Undeniability* says that nobody can produce, for the same accumulator value and the same element, both a membership proof and a non-membership proof that verify, even for a value that was not computed honestly from any set. This is what FENCE needs, because the accumulator is computed by the model owner, the party that the refusers do not trust. *Indistinguishability* says that the accumulator value and the non-membership proofs reveal nothing about which set was accumulated. In FENCE, the accumulated set is the training set and the queries are refusal lists, so this property is what lets a model owner prove exclusion without disclosing its data.

A cryptographic accumulator can be instantiated in several ways. For instance, accumulators with non-membership proofs can be built from hash trees [8]. The more widely used RSA accumulator was introduced by Benaloh and de Mare [5] and later extended to support non-membership proofs [37]. FENCE uses the accumulator only through its three properties (correctness, undeniability, and indistinguishability), so any accumulator that has them can be used.

4.3 Proof of the Model Description

In our verifiable exclusion scheme, we require that the model owner first publish a description of its model. This description includes a short commitment to the tokens in its training data, a commitment to its training algorithm and to any model it starts from, a fingerprint of its weights, and a proof that this description is correct. The proof is what ensures that the model was really trained on the committed data with the committed training algorithm and public randomness.

There are several ways to realize this proof in line with our scheme, and we describe three routes based on prior work. Proof-of-Learning [31] records checkpoints and batches during training, which is relatively cheap. Another route is zero-knowledge proofs of training [1, 27], which give cryptographic soundness at a cost that is currently practical only for small models. Hardware attestation of the training environment, for example with trusted execution environments [13], is a third route, with trust placed in the hardware vendor.

4.4 FENCE Workflow

We now combine these building blocks into the FENCE workflow: tokens, which represent features of data items (Section 4.1); the cryptographic accumulator (Section 4.2); and a proof that the model description is correct (Section 4.3).

Setup. The setup authority generates the accumulator's parameters and fixes how tokens are computed, that is, which features are compared and which hash function is used. This happens once, the authority keeps no secret, and the result is shared by all model owners, refusers, and registries.

Register refusals. The refuser, or a registry on their behalf, computes the tokens of what is to be protected, and the registry keeps the resulting refusal list R. A collective's list is the union of its members' tokens, so individual and collective refusal use the same mechanism.

Commit to the training data. Before training, the model owner computes the tokens of its training data, accumulates them into a single short commitment using a cryptographic accumulator, and publishes it together with a commitment to its training algorithm and any starting weights.

Train and attest. Only after these commitments are published is the training randomness read from the public source. The model owner uses this randomness to train its model, publishes the model or a fingerprint of its weights, and attaches an attestation proving that the model is the output of the committed algorithm, run from the committed starting weights with the public randomness on exactly the committed training set. Together, these form the published model description.

Prove exclusion. When a refuser or registry asks about a refusal list, the model owner checks it against its committed tokens. If any token on the list is also a token of the training data, no proof exists, and the owner has to remove the offending items, commit again, and retrain. Otherwise the owner produces one Proof of Exclusion for the whole list.

Verify exclusion. Anyone who holds the refusal list can check the model owner's Proof of Exclusion. The verifier first checks the attestation, which needs to be done only once per model, and then checks the Proof of Exclusion against the published commitment and the refusal list.

Refusal lists inherently grow. Thus, a proof can be made up against a specific, dated version of a list, and an owner then extends its coverage by publishing a further proof for the tokens added since. In practice this could be per training run or at fixed intervals.

4.5 What FENCE Guarantees

Section 3 stated three guarantees that any verifiable exclusion scheme must provide. Here we explain why FENCE provides each of them, and which property of its building blocks (the hash function, the accumulator of Section 4.2, and the attestation of Section 4.3) each one relies on. Each paragraph is an informal version of a theorem.

Completeness. A model owner that does not use refused data can always prove it, and no refuser can frame it. If none of the refused items are used, then by definition none of their features occurs in the training set. Because the hash function is collision resistant, different features practically never yield the same token, so none of the refused tokens is among the committed tokens either. The correctness of the accumulator then guarantees that the model owner can produce a proof of non-membership for the whole list, and that this proof is accepted. The same argument shows that a refuser cannot frame an honest owner: to block its proof, the refuser would have to find an item that shares a token with a training item without sharing a feature, which means finding a collision in the hash function.

Soundness. An accepted Proof of Exclusion guarantees that the published model was trained without the refused items. This rests on two building blocks. First, the proof of the model description binds the published model to the committed training set: any model owner that produces a valid proof must know a training set from which the model was trained and whose tokens are exactly the committed ones. In cryptographic terms, this training set can be extracted from the model owner; extraction is a thought experiment used in the security analysis to capture what the owner knows, not something a refuser or verifier can do, so it does not conflict with training-set privacy. Second, suppose a Proof of Exclusion were accepted even though a refused token is in that training set. The model owner, who knows the training set, could then also produce a proof that the token is in the commitment, and the same commitment would have both an accepted proof that the token is in it and an accepted proof that it is not. The undeniability of the accumulator rules this out, even

when the commitment was computed dishonestly. This is an important guarantee of FENCE, as the model owner is the party in power, and the one the refusers do not trust.

Training-set privacy. Training-set privacy protects the model owner's training set from being leaked. It follows directly from the indistinguishability of the accumulator: the commitment is an accumulator of the training tokens and a Proof of Exclusion is a proof of non-membership, so anything a refuser could learn about the training set from them would also distinguish accumulated sets, which indistinguishability rules out. The only thing revealed is whether the model owner can produce a Proof of Exclusion for a given refusal list. If the list shares a token with the training set, no such proof exists, and the owner's failure to produce one reveals that the training set uses a refused item; this is exactly what the scheme is meant to reveal.

4.6 Proactive vs. Retroactive Deployment

FENCE is used in two situations as a way to protect a user's rights to their data: proactive refusal and retroactive removal.

In proactive refusal, the refusal list exists before training. The owner excludes listed items, commits, trains, and provides the Proof of Exclusion on request. Nothing has to be unlearned because the data was never included, and the proof of exclusion is exact, where unlearning is approximate. Here, the boundary is established before any trespass occurs.

In the retroactive scenario, the model owner must: (1) remove the data from the training set, (2) commit to a new training set, (3) retrain the model and attest the new training run, (4) generate a new Proof of Exclusion for the new model. Step 3 remains expensive and is part of the same challenge unlearning addresses. FENCE does not eliminate this cost. What it adds is a verifiable guarantee that the data was actually removed from the training pipeline before retraining, which approximate unlearning methods do not provide.

4.7 Computational Costs

FENCE can be built from any universal accumulator with the three properties of Section 4.2; we use the RSA instantiation to give asymptotic costs, summarized in Table 1 (Appendix C). The model owner computes the commitment once per training run, with one modular exponentiation per training token, and producing a Proof of Exclusion costs about as much again, since the proof depends on every token in the committed set. Because a registry can collect tokens and submit them as one list, the owner pays this cost once per list rather than once per token, and with succinct proofs the proof shrinks to a constant number of group elements that can be checked with a constant number of exponentiations [7]. Verifying a proof for a single token takes only two modular exponentiations, independent of the size of the training set, so a refuser can check a proof on any device, including a phone. These costs exclude the training attestation, whose cost depends on the route chosen (Section 4.3). Two points matter here. First, the cost falls on the model owner, who uses other people's data in the first place, has far more computational power than any refuser, and is the party in power: for a training set of billions of items, computing the commitment and each proof is substantial, but it grows only linearly with the training set, as training itself does, while a refuser only computes hashes and checks a short proof. Second, the purpose of FENCE is to show that exclusion from a training set can be proven at all. As current disputes show, data use is being litigated while companies have no way to prove anything; with FENCE, a model owner can no longer claim that verification is impossible, which shifts the policy conversation from "this cannot be done" to who should be required to do it.

5 Discussion on FENCE's legal operation

Although the FENCE might presuppose a certain regulatory landscape, it does not enforce one. Its legal integration involves decisions that vary with jurisdiction, legal instrument, and data landscape.

In our background (Section 2.2), we argue that FENCE may find particularly salient fit in copyright law and privacy law. Proactive verifiable exclusion disproves claims of data misappropriation, while retroactive deletion provides compliance evidence. We also note that FENCE's construction is well-suited to domains where sensitive data comes in discrete, individually attributable records.

In this section, we discuss legal provisions that would support FENCE's deployment, and provide an evaluation of FENCE's hypothetical deployment in two legal landscapes. We finish by reflecting on cryptography and law's intersection.

5.1 Proactive refusal in the EU: a machine-readable opt-out registry

Article 4(3) of the EU's Copyright in the Digital Single Market Directive (DSM), introduced in 2021, affords individuals the right to declare their public online data off-limits to AI training provided that they opt out "in an appropriate manner, such as machine-readable means" [22]. Article 53(1)(c) of the EU's Artificial Intelligence Act mandates that AI companies themselves proactively demonstrate compliance of copyright and privacy laws by putting into place a compliance policy—including using "state-of-the-art technologies" to comply with the reservation of rights expressed in Article 4(3).

Notable is the lack of a prescriptive technological solution. The EU does not offer a dictionary definition for "appropriate" machine-readable means, though the Commission aims to publish a list of generally agreed-upon opt-out protocols [14, 20], reviewed biennially alongside the GPAI Code of Practice, against which the AI Office will assess compliance [20, 23]. Model providers can demonstrate compliance by adhering to the Code's copyright chapter [19]. Similarly, "state-of-the-art technologies" advises a dynamic bar that rises with innovations in the affordances and practicality of compliance technology.

5.1.1 How does FENCE *fit in?* Altogether, the EU's AI law adopts a regulatory posture that places the onus on right-holders to opt out proactively, and onus on model providers to produce a proactive, demonstrable compliance policy. This is in line with the method of *proactive refusal* FENCE affords.

Governance is at an inflection point where best-practice technological approaches are still being discussed by lawmakers and stakeholders, and coordination across stakeholders is difficult. A recent Commission-funded feasibility study, in addressing these uncertainties, proposes a hybrid opt-out registry: a central registry holds content fingerprints of opted-out works, encoded with the ISO International Standard Content Code (ISCC). Distributed onboarding operators—distributed across sectors—populate the registry. The registry would complement but not replace existing, sector-specific opt-out procedures and source-based signals such as robots.txt. FENCE *slots naturally into this architecture rather than competing with it.*

Parties. The study's registry plays the role of FENCE's **registry**, and sector-specific operators may populate it on behalf of refusers. As parameter generation requires a trusted party, the registry's operator is a good fit. However, sector-specific stakeholders should have input in deciding on hashed features. FENCE allows anybody to verify a proof in milliseconds, but regular **verifiers** may plausibly be the centralized registry (which could run batch verifications each time a model provider submits a new commitment and proof) and the EU AI Office (which audits Article 53(1)(c) compliance). Sector nodes could also verify on behalf of the rightholders they service. Crucially, FENCE allows trust to be distributed.

Opt-out expressions. The study's registry supports two kinds of opt-out signals: *source-based* (website- or platform-wide opt-out) and *work-based* (opt-out tied to an individual work). In FENCE, this maps directly to our proposed source-based and work-based opt-out tokens (Section 4.1). The ISCC—a standardized way of deriving a fingerprint from work content—can directly template the hashed features, without change to the FENCE protocol.

Verification. The study's registry design improves the expression and detection of opt-out signals, but compliance remains self-reported. FENCE allows proactive and retroactive proofs of exclusion (Section 4.6) to be demonstrated on dated versions of a registry list. Its training attestation allows training set commitment to be bound to the released model, but even if attestation is not perfectly robust, the published commitment allows a verifiable, privacy-preserving, and falsifiable claim to be brought to the public, where none such existed before.

A new legal mechanism: shifting the burden of production. A cryptographic registry may be in place, but what—beyond the prospect of a shorter legal dispute—motivates AI providers to generate expensive Proofs of Exclusion? A legal incentive is necessary, and we propose a regulation that explicitly shifts the burden of production (Section 2.1) to the model provider if they are accused of data misappropriation but fail to produce exclusion evidence. Specifically, model owners are afforded a *rebuttable presumption of compliance* if they provide a valid, correctly-dated Proof of Exclusion. In this case, claimants must prove independently that data has been ingested into the model to reverse the presumption. Conversely, model owners trigger a *rebuttable presumption of use* if they fail to provide a Proof of Exclusion. This means they are presumed to have used the refused data unless they convincingly prove otherwise. FENCE allows model owners to produce a Proof of Exclusion while preserving dataset privacy, making such an approach feasible for the first time. Our design draws from the EU's Product Liability Directive, described in Section 2.1, which recognizes the information asymmetry between claimants and vendors of proprietary products that have allegedly caused harm, and uses vendor non-compliance as a trigger for a shift in burden of production.

On the model provider side, FENCE's accumulator architecture means that proofs are cheaper to generate per-token if tokens are in larger batches (Section 4.7). These factors—combined with the feasibility study's own recommendations about sector-level liaisons—lead us to argue that in the copyright case, FENCE is *most realistically deployed at a sectoral level, with sectoral enforcement.*

5.2 Retroactive refusal in the US: rectifying privacy violations

In the US, there is no federal regime empowering individuals to reserve data from training sets, nor a comprehensive privacy law. Generally, the removal of data from training sets is ordered *after* a court rules that a trained model has infringed on individuals' privacy rights. The US is thus a natural setting for FENCE's *retroactive removal*, though its *proactive exclusion* is also in demand, as regimes ordering remedial data deletion generally take measures to ensure future compliance.

A case study. We return to *FTC v. Amazon*—the case outlined in our introduction—where Amazon illegally retained children's voice recordings for algorithm training, and its compliance with the FTC's remedial deletion order rested on little more than its own word. With cryptography, enforcers gain a powerful tool to audit compliance—but how exactly could FENCE come in?

Parties. Refusal tokens come as *source tokens* for the inactive children accounts, and *content tokens* for audio recordings. There is no standing registry, so the enforcing authority—here, the FTC—assumes the role of FENCE's **registry** and **verifier**, such that it can verify Amazon's compliance moving forwards. The **setup authority's** roles may be split: the FTC decides on features relevant to the removal order's contents, while an appointed third party should generate the accumulator

parameters, to avoid either party being able to forge proofs. The refusal registry is maintained such that, in future training runs of Amazon's offending model, the FTC can order Amazon to generate training set commitments and a single, batched **Proof of Exclusion** that demonstrates they have not reused violating data. FENCE thus enables forward prohibition to be audited routinely, automatically, and privately. A one-time human audit is complex and costly to the enforcement authority, but FENCE shifts the audit cost to the company (via computationally expensive proof generation), and ensures that the task of verification can be done by anybody.

Legal mechanism: proof as an order obligation. Unlike the EU mechanism, no new legislation is required to enforce FENCE. FTC orders routinely specify how compliance is reported, so the FTC could simply order periodic Proofs of Exclusion against an enforcer-held list instead of demanding that Amazon swear in writing that it has removed the data. Failure to provide the proof, an order violation, can be penalized without presumption about data use.

In reflecting on FENCE's hypothetical deployment within a US privacy regime, we put forward a few reflections. First, FENCE's *retroactive exclusion* is helpful in verifying that an offending model has since been retrained, but it is less equipped to demonstrate without doubt that the original offending dataset has been deleted. This difficulty is present with any kind of audit that lacks full transparency, but FENCE arguably supports a remedial approach that has more practical impact. Second, the patchwork nature of US privacy laws presents a difficulty in prescribing a general way of deploying FENCE. The powers the FTC has in auditing and demanding data exclusion, for instance, may vary from that of the BIPA. It must also be noted that, if a company's training set is found violating by multiple regimes—each with tokens incorporating different features—it may accumulate a hefty burden of proof generation.

5.3 Reflection on FENCE & the law

Our two case studies give us insight into how FENCE, on one hand, may push resonant policymaking further by creating unexpected legal affordances—but on another, is still heavily constrained by regulatory landscape.

Reflecting at the intersection of computer science and law, we recall Rogaway's still-urgent reminder that "cryptography rearranges power; it configures who can do what, from what." [41]. Resonating with that ethos, FENCE's design explicitly embodies a certain politics of evidentiary burden. As we contemplate its diverse deployment possibilities, however, we are reminded that, while a technology may presume a certain politics, it doesn't implement those politics itself. An integrated and co-constructive approach is needed.

But cryptography and lawmaking are both uncommonly *exacting* in how they formalize a rearrangement in power—whether through protocol specifications or through precise statutory text: they compel us to be especially thoughtful about how an intervention can and should persist into the future.

6 Conclusion

This paper has introduced FENCE (Framework for Exclusion via Non-membership Cryptographic Evidence), a cryptographic framework that transforms data sovereignty from a trust-based process into a mathematically verifiable one. Just as a physical fence establishes a visible, enforceable boundary, FENCE allows individuals and communities to put up a cryptographic boundary around their data that model owners cannot quietly cross. By combining cryptographic rigor with sociotechnical awareness, deployment considerations for real users, and policy analysis grounded in existing regulatory frameworks, FENCE demonstrates that verifiable data refusal is not merely a theoretical possibility but a practical and urgent contribution.

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A Cryptographic Accumulator

In this section, we provide the formal definitions and syntax of a universal cryptographic accumulator [3, 38]. Throughout, λ denotes the security parameter, negl a negligible function, and PPT stands for probabilistic polynomial time.

Definition A.1 (Cryptographic accumulator). A cryptographic accumulator consists of six PPT algorithms:

- $\mathsf{Setup}(1^\lambda) \to \mathsf{pp}$, which also fixes an efficiently recognisable domain $\mathcal{D}_{\mathsf{pp}}$.
- $\mathsf{Acc}(\mathsf{pp}, S) \to (A, \mathsf{aux})$ for a finite $S \subseteq \mathcal{D}_{\mathsf{pp}}$.
- $\mathsf{MemProve}(\mathsf{pp}, \mathsf{aux}, x) \to w$ for $x \in S$.
- $\mathsf{MemVerify}(\mathsf{pp}, A, x, w) \to \{0, 1\}$, deterministic.
- $\mathsf{NonMemProve}(\mathsf{pp}, \mathsf{aux}, X) \to \bar{w}$ for finite non-empty $X \subseteq \mathcal{D}_{\mathsf{pp}}$ with $X \cap S = \emptyset$; it outputs $\perp$ if $X \cap S \neq \emptyset$.
- $\mathsf{NonMemVerify}(\mathsf{pp}, A, X, \bar{w}) \to \{0, 1\}$, deterministic.

We use three security properties of a cryptographic accumulator: *correctness, undeniability, and indistinguishability*. We do not state collision-freeness as a separate definition, as it is implied by undeniability [17].

Correctness says that the scheme works for honest parties. If the accumulator was computed honestly from a set, then every element of the set has a membership witness that verifies, and every collection of elements outside the set has a non-membership witness that verifies. The negligible error allows for accumulators that use randomness. In FENCE, correctness is what lets a model owner whose training set avoids a refusal list always produce an accepting proof (Definition A.2).

Definition A.2 (Correctness). A universal accumulator is *correct* if for all λ, all $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda)$, all finite $S \subseteq \mathcal{D}_{\mathsf{pp}}$, every $x \in S$, and every finite non-empty $X \subseteq \mathcal{D}_{\mathsf{pp}}$ with $X \cap S = \emptyset$, the following

hold with probability at least $1 - \mathsf{negl}(\lambda)$ over $(A, \mathsf{aux}) \leftarrow \mathsf{Acc}(\mathsf{pp}, S)$:

$$\mathsf{MemVerify}\big(\mathsf{pp}, A, x, \mathsf{MemProve}(\mathsf{pp}, \mathsf{aux}, x)\big) = 1,$$
$$\mathsf{NonMemVerify}\big(\mathsf{pp}, A, X, \mathsf{NonMemProve}(\mathsf{pp}, \mathsf{aux}, X)\big) = 1.$$

Undeniability says that nobody can produce, for the same accumulator value and the same element, both a membership witness and a non-membership witness. The adversary chooses the accumulator value itself, so the property holds even for a value that was not computed honestly from any set. This is what FENCE needs, because the accumulator is computed by the model owner, the party that the refusers do not trust. An accumulator value hides its set, so an element being "in" a value the owner chose can only mean that a membership witness for it exists. Undeniability guarantees that such a witness and an accepted non-membership proof for the same element can never both exist (Definition A.3).

Definition A.3 (Undeniability). A universal accumulator is *undeniable* if for every PPT adversary $\mathcal{A}$,

$$\Pr\left[\begin{array}{l} \mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda);\ (A, X, \pi, x, w) \leftarrow \mathcal{A}(\mathsf{pp}) : \\ x \in X \ \wedge\ \mathsf{NonMemVerify}(\mathsf{pp}, A, X, \pi) = 1 \ \wedge\ \mathsf{MemVerify}(\mathsf{pp}, A, x, w) = 1 \end{array}\right] \le \mathsf{negl}(\lambda).$$

Indistinguishability states that the accumulator value and non-membership witnesses reveal nothing about which set was accumulated. The adversary may pick two candidate sets, receive the accumulator of one of them, and request non-membership witnesses for elements that lie in neither set; it still cannot tell which set was used any better than by guessing. Queries for elements in one set but not the other are excluded, because a witness would exist in one case and not in the other, which would reveal the answer trivially. Our definition adapts that of Derler, Hanser, and Slamanig [17]: the adversary receives non-membership witnesses but no membership witnesses, and the number of distinct elements it may query is bounded. In FENCE, the accumulated set is the training set and the queries are refusal lists, so this property is what lets a model owner prove exclusion without disclosing its data (Definition A.4).

Definition A.4 (Indistinguishability). Let $\kappa \ge 1$. A universal accumulator is *κ-bounded indistinguishable* if for every PPT adversary $\mathcal{A}$,

$$\Pr\left[\begin{array}{l} \mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda);\ (S_0, S_1, \sigma) \leftarrow \mathcal{A}(\mathsf{pp});\ b \leftarrow \{0,1\}; \\ (A, \mathsf{aux}) \leftarrow \mathsf{Acc}(\mathsf{pp}, S_b);\ b' \leftarrow \mathcal{A}^{\mathcal{O}}(\sigma, A) :\ b' = b \end{array}\right] \le \frac{1}{2} + \mathsf{negl}(\lambda),$$

where $S_0, S_1 \subseteq \mathcal{D}_{\mathsf{pp}}$ are finite, and $\mathcal{O}$, on input a finite non-empty $X \subseteq \mathcal{D}_{\mathsf{pp}}$ with $X \cap (S_0 \cup S_1) = \emptyset$, returns $\mathsf{NonMemProve}(\mathsf{pp}, \mathsf{aux}, X)$, with at most κ distinct elements submitted across all queries.

B FENCE Security Definitions

We state the guarantees of Section 3 formally but do not give formal proofs. Each follows by a direct reduction to one property of the building blocks, as sketched in Section **??**: completeness from the correctness of the accumulator and the collision resistance of H, soundness from attested training and the undeniability of the accumulator, and training-set privacy from the indistinguishability of the accumulator.

We use the following notation. T is a training set, a finite set of data items, and D is a set of refused items. For an item d, $\mathsf{Feat}(d)$ is its set of features and $\mathsf{Tok}(d) = \{H(f) : f \in \mathsf{Feat}(d)\}$ its set of tokens; for a set of items T, we write $\mathsf{Feat}(T)$ and $\mathsf{Tok}(T)$ for the unions over its items. A refusal list R is a set of tokens, and the refusal list for D is $R = \mathsf{Tok}(D)$. FENCE.Setup outputs public parameters pp, which include the accumulator's parameters and the hash function H. $\mathsf{Commit}(\mathsf{pp}, T)$ outputs a commitment A to $\mathsf{Tok}(T)$ and a private state st kept by the model owner. $\mathsf{ProveExcl}(\mathsf{pp}, \mathsf{st}, R)$ outputs

a Proof of Exclusion π for a refusal list R, which anyone can check with $\mathsf{VerifyExcl}(\mathsf{pp}, A, R, \pi)$, and ProveIncl and VerifyIncl do the same for a single token in the commitment.

Completeness lets the adversary choose both the training set and the refused items, so it also rules out framing: registering items that share no feature with the training set cannot cause an honest model owner's proof to fail.

Definition B.1 (Completeness). FENCE is *complete* if for every PPT adversary $\mathcal{A}$,

$$\Pr\left[\begin{array}{l}\mathsf{pp} \leftarrow \mathsf{FENCE.Setup}(1^\lambda);\ (T, D) \leftarrow \mathcal{A}(\mathsf{pp});\ R := \mathsf{Tok}(D);\\ (A, \mathsf{st}) \leftarrow \mathsf{Commit}(\mathsf{pp}, T);\ \pi \leftarrow \mathsf{ProveExcl}(\mathsf{pp}, \mathsf{st}, R) :\\ \mathsf{Feat}(T) \cap \mathsf{Feat}(D) = \emptyset \ \wedge\ \mathsf{VerifyExcl}(\mathsf{pp}, A, R, \pi) = 0\end{array}\right] \leq \mathsf{negl}(\lambda).$$

Exclusion soundness says that a model owner cannot claim that a token is both outside and inside its commitment. The adversary plays a dishonest model owner and chooses everything itself, including the commitment A, which need not have been computed honestly from any training set. If it wins it produces a Proof of Exclusion π for a refusal list R that is accepted, together with accepted evidence w that some token r on that list is in the same commitment. In FENCE, this means that once a Proof of Exclusion for a refused token has been accepted, the model owner can never afterwards show that the token belongs to its committed training set. For instance to an auditor who inspects the training data and asks for inclusion evidence item by item. Exclusion soundness is a statement about the commitment alone and combined with attested training, it yields the soundness guarantee of our verifiable exclusion scheme.

Definition B.2 (Exclusion soundness). FENCE is *exclusion sound* if for every PPT adversary $\mathcal{A}$,

$$\Pr\left[\begin{array}{l}\mathsf{pp} \leftarrow \mathsf{FENCE.Setup}(1^\lambda);\ (A, R, \pi, r, w) \leftarrow \mathcal{A}(\mathsf{pp}) :\\ r \in R \ \wedge\ \mathsf{VerifyExcl}(\mathsf{pp}, A, R, \pi) = 1 \ \wedge\ \mathsf{VerifyIncl}(\mathsf{pp}, A, r, w) = 1\end{array}\right] \leq \mathsf{negl}(\lambda).$$

Training-set privacy says that the commitment and the Proofs of Exclusion reveal nothing about the model owner's training set beyond the answers that refusers are entitled to. The adversary plays a curious refuser, or anyone else who sees the proofs. It chooses the two candidate training sets T_0 and T_1, receives the commitment to one of them, and then may request Proofs of Exclusion for refusal lists of its choice. It wins if it can tell which training set was committed to any better than by guessing. Refusal lists may only contain tokens that appear in neither training set, because for a token in one set but not the other, a proof would exist in one case and not in the other, revealing the answer trivially.

Definition B.3 (Training-set privacy). FENCE has *κ-bounded training-set privacy* if for every PPT adversary $\mathcal{A}$,

$$\Pr\left[\begin{array}{l}\mathsf{pp} \leftarrow \mathsf{FENCE.Setup}(1^\lambda);\ (T_0, T_1, \sigma) \leftarrow \mathcal{A}(\mathsf{pp});\ b \leftarrow \{0,1\};\\ (A, \mathsf{st}) \leftarrow \mathsf{Commit}(\mathsf{pp}, T_b);\ b' \leftarrow \mathcal{A}^{\mathcal{O}}(\sigma, A) :\ b' = b\end{array}\right] \leq \frac{1}{2} + \mathsf{negl}(\lambda),$$

where $\mathcal{O}$ is an oracle that, on input a refusal list R with $R \cap \big(\mathsf{Tok}(T_0) \cup \mathsf{Tok}(T_1)\big) = \emptyset$, returns $\mathsf{ProveExcl}(\mathsf{pp}, \mathsf{st}, R)$, with at most κ distinct tokens queried in total.

C FENCE Asymptotic Costs

Table 1 summarizes the asymptotic costs of FENCE for each party when it is instantiated with the RSA accumulator, as discussed in Section 4.7. The model owner's costs grow with the size of its training set, while a refuser only computes hashes and a verifier performs a constant number of exponentiations per token, or a constant number overall with succinct proofs. The table covers only the steps that FENCE adds; the cost of the training attestation depends on the route chosen (Section 4.3) and is not included.

Operation	**Refuser**	**Model owner**	**Verifier**	**Size**
Tokenize k items	$O(k)$ hashes	–	–	k tokens
Commit to training set	–	$O(m)$ exp.	–	1 group elt.
Proof of Exclusion (k tokens)	–	$O(m+k)$ exp.	–	$O(k)$ bits (plain) $O(1)$ (succinct)
Verify, one token	–	–	2 exp.	–
Verify, k tokens	–	–	$O(k)$ mults (plain) $O(1)$ exp. (succinct)	–
Remove an item	–	as Commit	–	new commitment

Table 1. Asymptotic costs of FENCE with the RSA instantiation. m: training tokens; k: refusal tokens; exp.: one modular exponentiation with a token as exponent. We include plain proofs and also succinct proofs of Boneh, Bünz and Fisch [7], and their verification also reads the list of k tokens once.